「Prove that if $\{a_n\}$ is sequence of integers s.t infinitely many non-zero term, then the convergence bound of $\sum a_n$ is at most 1.

Sol). If $a_n = 1$ $\forall n \in \mathbb{N}$, then $R = 1$.

And, by the Theorem, $R > 1 \iff \limsup_{n \to \infty} \sqrt[n]{|a_n|} < 1$.

Put $\limsup \sqrt[n]{a_n} = \beta < 1$. Then, for $\beta < \alpha < 1$,

$$\exists N \in \mathbb{N} \text{ s.t } n \geq N \Rightarrow \sqrt[n]{|a_n|} < \alpha$$

$$\iff |a_n| < \alpha^n < 1$$

Contradiction.

Section 3.11.

Let $a_n > 0$, $S_n = a_1 + \cdots + a_n$, and $\sum a_n$ diverges. Prove that

a) $\sum \frac{a_n}{1+a_n}$ diverges.

b) $\sum \frac{a_n}{S_n}$ diverges.

c) $\sum \frac{a_n}{S_n^2}$ converges.

d) $\sum \frac{a_n}{1+na_n}$ and $\sum \frac{a_n}{1+n^2 a_n}$ converges?

Sol a) Suppose that $\limsup_{n\to\infty} a_n < 1$. Then, there exists $N \in \mathbb{N}$ s.t

$$n \geq N \Rightarrow a_n \leq 1$$

$a_n < \alpha + \varepsilon$

Meanwhile, if $a_n \leq 1$,

$\alpha - \varepsilon < a_n$

$$\frac{a_n}{2} \leq \frac{a_n}{1+a_n}$$

Therefore $\sum \frac{a_n}{1+a_n}$ diverges.

Suppose that $\limsup_{n\to\infty} a_n \geq 1$. Then, there exist infinitely many terms larger than $\frac{1}{2}$.

Since $a_n \geq \frac{1}{2}$ implies

$$a_n \geq \frac{1}{2} \Rightarrow 2a_n \geq 1 \Rightarrow \frac{1}{2a_n + a_n} \leq \frac{1}{1+a_n} \Rightarrow \frac{1}{3} \leq \frac{a_n}{1+a_n}$$

Therefore diverges.

Sol b). For any $N \in \mathbb{N}$ and $k \geq 0$,

$$\frac{a_{N+1}}{S_{N+1}} + \cdots + \frac{a_{N+k}}{S_{N+k}} \geq \frac{1}{S_{N+k}} \cdot (a_{N+1} + \cdots + a_{N+k}) = \frac{S_{N+k} - S_N}{S_{N+k}} = 1 - \frac{S_N}{S_{N+k}} \quad \cdots (*)$$

If $\sum \frac{a_n}{S_n}$ converges, then there exists $N \in \mathbb{N}$ s.t

$$m \geq n \geq N \Rightarrow \left| \sum_{k=n}^{m} \frac{a_n}{S_n} \right| < \frac{1}{2}.$$

But, since the inequality $(*)$, and $S_n = \sum_{i=0}^{n} a_i$ diverges, we can choose $k \geq N$ s.t

$$\frac{1}{2} \leq 1 - \frac{S_N}{S_{N+k}} \leq \sum_{i=N+1}^{N+k} \frac{a_i}{S_i}$$

which is contradiction.

Sol c) Since $S_{n-1} < S_n$ for any $n \in \mathbb{N}$,

$$\frac{a_n}{S_n^2} \le \frac{a_n}{S_{n-1}S_n} = \frac{S_n - S_{n-1}}{S_{n-1} \cdot S_n} = \frac{1}{S_{n-1}} - \frac{1}{S_n} \qquad (n \ge 2)$$

Now, for any $n \ge 2$,

$$\sum_{k=2}^{n} \frac{a_k}{S_k^2} \le \sum_{k=2}^{n} \frac{1}{S_{k-1}} - \frac{1}{S_k} = \frac{1}{S_1} - \frac{1}{S_n} \to \frac{1}{S_1} \quad \text{as } n \to \infty$$

Since $S_n$ diverges to $\infty$.

Sol d) First,

$$\frac{a_n}{1+n^2 a_n} \le \frac{a_n}{n^2 a_n} = \frac{1}{n^2}, \quad \text{therefore } \sum \frac{a_n}{1+n^2 a_n} \le \sum \frac{1}{n^2}, \text{ converges.}$$

To show the first series diverges,

If $\limsup_{n\to\infty} n a_n < 1$, then there exists $N \in \mathbb{N}$ s.t

$$n \ge N \Rightarrow n a_n < 1$$

which implies that

$$\frac{a_n}{2} \le \frac{a_n}{1+n a_n}$$

therefore $\sum \frac{a_n}{1+n a_n}$ diverges.

If $\limsup_{n\to\infty} n a_n \ge 1$, then there exists infinitely many $n \in \mathbb{N}$ s.t

$$n a_n \ge \frac{1}{2} \qquad 2 n a_n \ge 1$$

Which implies that

$$\frac{a_n}{1+n a_n} \ge \frac{a_n}{2 n a_n + n a_n} \ge \frac{1}{3n}$$

Therefore, $\sum \frac{a_n}{1+n a_n}$ diverges.

$$\frac{a_n}{1+n a_n} \ge \frac{a_n}{n + n a_n} = \frac{a_n}{n(1+a_n)}$$

Bernoui's

If $a > -1$, $(1+a)^n \ge 1 + na$.

Proof. $n = 1$, trivial. $(1+a = 1+a)$.

Suppose $n > 1$, stat. holds if $n$.

$$\begin{aligned}(1+a)^{n+1} &= (1+a)\cdot(1+a)^n \\ &\ge (1+a)\cdot(1+na) \\ &= (1 + na + a + na^2) \\ &\ge 1 + (n+1)a.\end{aligned}$$

Section 3, 12

Let $a_n > 0$, $\sum a_n$ converges, put $r_n = \sum_{m=n}^{\infty} a_m$

a). Prove $\forall N, K \in \mathbb{N}$,

$$\frac{a_N}{r_N} + \cdots + \frac{a_{N+K}}{r_{N+K}} > 1 - \frac{r_{N+K}}{r_N}$$

Deduce that $\sum \frac{a_n}{r_n}$ diverges.

$$\frac{r_N - a_N}{r_N} = \frac{r_{N+1}}{r_N}$$

b). Prove $\forall n \in \mathbb{N}$,

$$\frac{a_n}{\sqrt{r_n}} < 2(\sqrt{r_n} - \sqrt{r_{n+1}})$$

Deduce that $\sum \frac{a_n}{\sqrt{r_n}}$ converges.

Sol). Since $\sum a_n$ converges, put $A = \sum_{n=1}^{\infty} a_n$. Then,

$r_n \to 0$ ?

$$A - r_n = \sum_{k=1}^{n-1} a_k$$

And, $r_n$ is decreasing. Now, $r_N > r_{N+i} \Rightarrow \frac{1}{r_N} < \frac{1}{r_{N+i}}$ implies

$$\frac{a_N}{r_N} + \cdots + \frac{a_{N+K}}{r_{N+K}} > \frac{1}{r_N}(a_N + a_{N+1} + \cdots + a_{N+K})$$

$$= \frac{1}{r_N}(r_N - r_{N+K+1})$$

$$= 1 - \frac{r_{N+K+1}}{r_N} > 1 - \frac{r_{N+K}}{r_N}$$

Now, suppose $\sum \frac{a_n}{r_n}$ converges. Then, there exists $N \in \mathbb{N}$ s.t for all $K \in \mathbb{N}$,

$$\left| \sum_{n=N}^{N+K} \frac{a_n}{r_n} \right| < \frac{1}{2}.$$

Meanwhile, $\lim_{n\to\infty} r_n = 0$ because for any $\varepsilon > 0$, we can choose $N \in \mathbb{N}$ s.t

$$n \geq N \Rightarrow |r_n| = r_n = \sum_{m=n}^{\infty} r_m = A - \sum_{k=1}^{n-1} a_k < \varepsilon$$

Therefore, we can choose $R \in \mathbb{N}$ for fixed $N$,

$$K \geq R \Rightarrow \left| \sum_{n=N}^{N+K} \frac{a_n}{r_n} \right| > 1 - \frac{r_{N+K}}{r_N} \geq \frac{1}{2}.$$

It is contradiction.

Sol b). Since $a_n = r_n - r_{n+1} = (\sqrt{r_n} - \sqrt{r_{n+1}})(\sqrt{r_n} + \sqrt{r_{n+1}})$ $(n \geq 1)$

$$\frac{a_n}{\sqrt{r_n}} = (\sqrt{r_n} - \sqrt{r_{n+1}})\left(1 + \frac{\sqrt{r_{n+1}}}{\sqrt{r_n}}\right)$$

$$< (\sqrt{r_n} - \sqrt{r_{n+1}})(1 + 1) = 2(\sqrt{r_n} - \sqrt{r_{n+1}})$$

Therefore,

$$\sum_{k=1}^{n} \frac{a_k}{\sqrt{r_k}} < \sum_{k=1}^{n} 2(\sqrt{r_k} - \sqrt{r_{k+1}})$$

$$= 2(\sqrt{r_1} - \sqrt{r_{n+1}}) \to 2\sqrt{r_1} \quad \text{as } n \to \infty.$$

Thus $\sum \frac{a_n}{\sqrt{r_n}}$ converges.